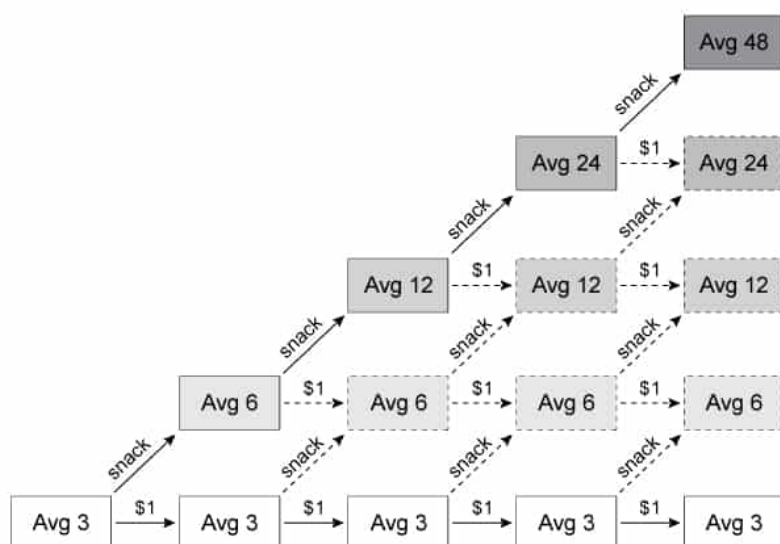


Researchers conducted a pair of studies to explore how rewards influence motivation and weight loss for those who are overweight or obese.

For Study 1, 20 overweight and 20 average-weight adults were invited to play a computer game consisting of a simple matching task in which participants earned a reward for getting multiple correct responses. Participants were allowed to select their own reward: Either \$1.00 or a 200-calorie vending machine snack (valued at \$1.00), such as a candy bar or bag of chips. For the first trial, participants were rewarded on an average of three correct responses (AVG3). Participants were then asked to choose between the food and monetary rewards. When a food reward was selected, the average number of correct responses required to earn the next reward doubled, but when a monetary reward was selected, the average number of correct responses required to earn the next reward remained the same (Figure 1). Results indicated that although there was no significant difference between the groups with regard to baseline self-reporting of hunger prior to the start of the computer tasks, overweight participants were five times more likely to choose the food rewards than average-weight participants.



## Figure 1 Study 1 reward schedule

For Study 2, obese participants were recruited for a 16-week weight-loss study and randomly assigned to one of two groups: A control group ( $n = 32$ ) or an experimental group ( $n = 35$ ). Control participants engaged in weekly weigh-ins only. Experimental participants engaged in weekly weigh-ins but also earned or lost money (put into an account and paid at the end of the study) based on their weigh-in goal, calculated each week based on current body mass index (BMI), age, and sex. Those who reached their weekly weight goal earned \$10; for each additional pound lost, participants earned another \$10, up to \$50/week. Participants in the experimental group who did not meet or exceed their weekly weight goal had \$20 deducted from their accounts. No significant differences were found in baseline variables for either group; however, members of the control group lost an average of 5.9 pounds, whereas members of the experimental group lost an average of 18.1 pounds.

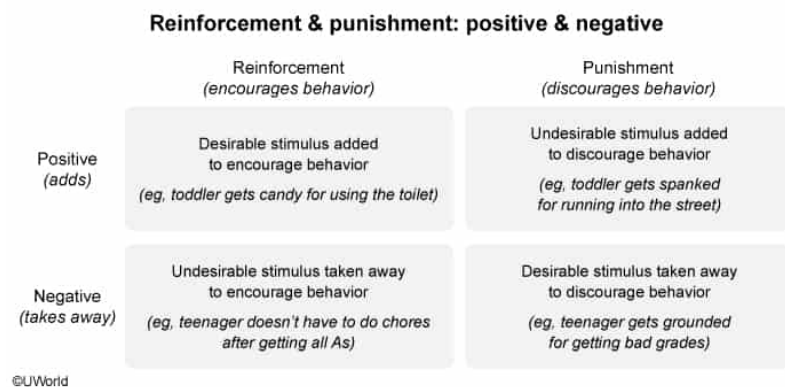
The experimental group in Study 2 was subjected to:

- ✔ ☒ A. positive reinforcement and negative punishment. [73%]  
☐ B. negative reinforcement and positive punishment. [1%]  
☐ C. positive reinforcement and positive punishment. [9%]  
☐ D. positive and negative reinforcement. [15%]

Correct

74% answered correctly

**Explanation:**



Operant conditioning is a type of associative learning whereby an organism learns to connect a behavior with an outcome or consequence, which is known as reinforcement (reward) or punishment (penalty).

Reinforcement encourages behavior. **Positive reinforcement** occurs when a **desirable stimulus** is **presented** to encourage a behavior. For example, a toddler receives a treat after successfully using the toilet. **Negative reinforcement** occurs when an **unpleasant stimulus** is **withdrawn** to encourage a behavior. For example, a teenager is given a week off from chores after getting good grades.

Punishment discourages behavior. **Positive punishment** occurs when an **unpleasant stimulus** is **applied** to discourage a behavior. For example, a toddler gets spanked for running into the street. **Negative punishment** occurs when a **desirable stimulus** is **withdrawn** to discourage a behavior. For example, a teenager loses freedom by being grounded for a week after getting bad grades.

In Study 2, earning money for meeting/exceeding a weight-loss goal is positive reinforcement, and taking money away for not meeting a weight-loss goal is negative punishment.

**(Choices B, C, and D)** There was no removal of an unpleasant

stimulus (negative reinforcement) or application of an unpleasant stimulus (positive punishment) in Study 2.

**Educational objective:**

Reinforcement encourages certain behaviors: Positive reinforcement provides a desirable stimulus (eg, receiving candy), whereas negative reinforcement takes away an undesirable stimulus (eg, removing chores). Punishment discourages certain behaviors: Positive punishment applies an undesirable stimulus (eg, spanking), whereas negative punishment takes away a desirable stimulus (eg, being grounded takes away freedom).

**Time spent:**0 sec

**Subject:**

Behavioral Sciences

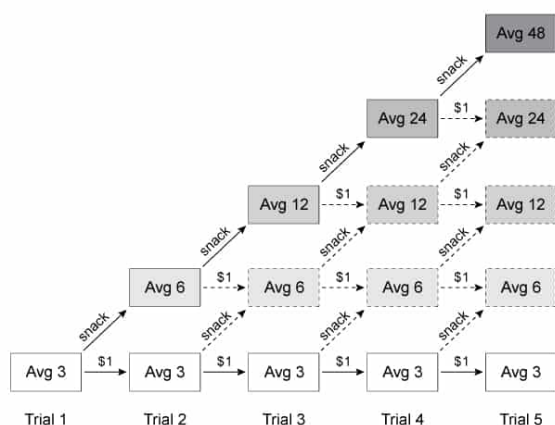
**QID:**400546

**System:**

Learning, Memory, and Cognition

Researchers conducted a pair of studies to explore how rewards influence motivation and weight loss for those who are overweight or obese.

For Study 1, 20 overweight and 20 average-weight adults were invited to play a computer game consisting of a simple matching task in which participants earned a reward for getting multiple correct responses. Participants were allowed to select their own reward: Either \$1.00 or a 200-calorie vending machine snack (valued at \$1.00), such as a candy bar or bag of chips. For the first trial, participants were rewarded on an average of three correct responses (AVG3). Participants were then asked to choose between the food and monetary rewards. When a food reward was selected, the average number of correct responses required to earn the next reward doubled, but when a monetary reward was selected, the average number of correct responses required to earn the next reward remained the same (Figure 1). Results indicated that although there was no significant difference between the groups with regard to baseline self-reporting of hunger prior to the start of the computer tasks, overweight participants were five times more likely to choose the food rewards than average-weight participants.



**Figure 1** Study 1 reward schedule

For Study 2, obese participants were recruited for a 16-week weight-loss study and randomly assigned to one of two groups: A control group ( $n = 32$ ) or an experimental group ( $n = 35$ ). Control participants engaged in weekly weigh-ins only. Experimental participants engaged in weekly weigh-ins but also earned or lost money (put into an account and paid at the end of the study) based on their weigh-in goal, calculated each week based on current body mass index (BMI), age, and sex. Those who reached their weekly weight goal earned \$10; for each additional pound lost, participants earned another \$10, up to \$50/week. Participants in the experimental group who did not meet or exceed their weekly weight goal had \$20 deducted from their accounts. No significant differences were found in baseline variables for either group; however, members of the control group lost an average of 5.9 pounds, whereas members of the experimental group lost an average of 18.1 pounds.

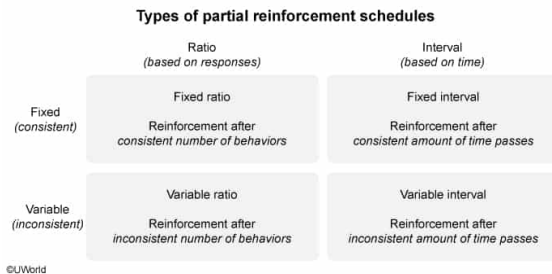
Which of the following most accurately reflects the reinforcement schedules used by Study 1 and the experimental group in Study 2, respectively?

- ☐ A. Variable ratio; variable interval [3%]
- ✓ ☒ B. Variable ratio; fixed interval [51%]
- ☐ C. Fixed ratio; variable interval [8%]
- ☐ D. Fixed ratio; fixed interval [36%]

Correct

51% answered correctly

## Explanation:



Schedules are used in operant conditioning to train and/or maintain learned behaviors through reinforcement or punishment. Continuous schedules provide a reward or punishment after every behavior, which produces the fastest learning using reinforcement and is the most effective way to stop unwanted behavior using punishment.

Partial **reinforcement schedules** are most often used to **maintain learned behaviors**, and include:

- **Fixed ratio:** Behavior rewarded after a consistent number of responses. For example, a customer loyalty program awards a free coffee after ten coffee purchases.
- **Variable ratio:** Behavior rewarded after an inconsistent number of responses, usually after some average number of responses, as in Study 1. The AVG3 schedule might provide a reward after four correct responses, then after two, then after three, and so on.
- **Fixed interval:** Behavior rewarded after a consistent amount of time, regardless of how many behaviors have occurred. For example, employees who are paid a weekly salary receive the same amount of money each Friday regardless of how hard they have worked during the week.
- **Variable interval:** Behavior rewarded after an inconsistent amount of time, regardless of how many behaviors have occurred. For example, when awaiting an email, the frequency with which users refresh their in-boxes does not affect when a new email arrives.

The reinforcement schedule used by Study 1 was a variable ratio schedule, because the reward was provided after some average number of correct responses. The reinforcement schedule used for the experimental group in Study 2 was a fixed interval schedule, because rewards occurred on a weekly basis.

**(Choice A)** A variable interval schedule provides rewards after unpredictable time increments. Study 2 provided reinforcement (money) on a fixed time schedule.

**(Choices C and D)** A fixed ratio schedule provides a reward for a set number of correct responses. Study 1 provided rewards after an average number of correct responses, consistent with a variable ratio schedule.

**Educational objective:**

Reinforcement schedules can be continuous (reward given every time) or partial. Partial schedules include fixed or variable ratio (reward given after consistent or inconsistent number of behaviors) and fixed or variable interval (reward given after consistent or inconsistent amount of time).

**Time spent:**0 sec

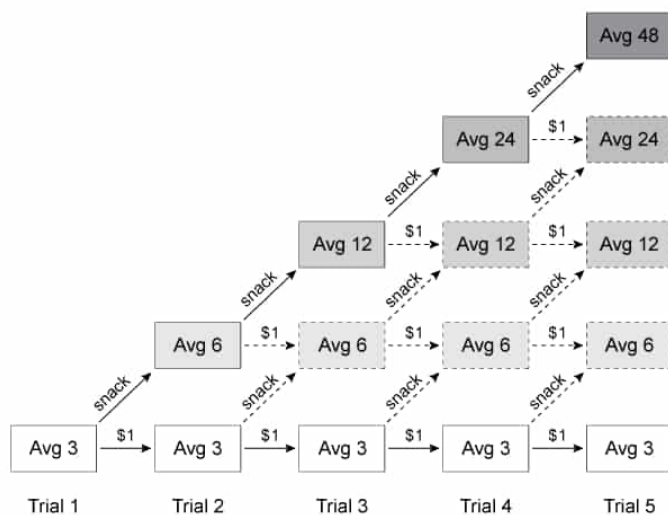
**Subject:**  
Behavioral Sciences

**QID:**400547

**System:**  
Learning, Memory, and Cognition

Researchers conducted a pair of studies to explore how rewards influence motivation and weight loss for those who are overweight or obese.

For Study 1, 20 overweight and 20 average-weight adults were invited to play a computer game consisting of a simple matching task in which participants earned a reward for getting multiple correct responses. Participants were allowed to select their own reward: Either \$1.00 or a 200-calorie vending machine snack (valued at \$1.00), such as a candy bar or bag of chips. For the first trial, participants were rewarded on an average of three correct responses (AVG3). Participants were then asked to choose between the food and monetary rewards. When a food reward was selected, the average number of correct responses required to earn the next reward doubled, but when a monetary reward was selected, the average number of correct responses required to earn the next reward remained the same (Figure 1). Results indicated that although there was no significant difference between the groups with regard to baseline self-reporting of hunger prior to the start of the computer tasks, overweight participants were five times more likely to choose the food rewards than average-weight participants.



**Figure 1** Study 1 reward schedule

For Study 2, obese participants were recruited for a 16-week weight-loss study and randomly assigned to one of two groups: A control group ( $n = 32$ ) or an experimental group ( $n = 35$ ). Control participants engaged in weekly weigh-ins only. Experimental participants engaged in weekly weigh-ins but also earned or lost money (put into an account and paid at



the end of the study) based on their weigh-in goal, calculated each week based on current body mass index (BMI), age, and sex. Those who reached their weekly weight goal earned \$10; for each additional pound lost, participants earned another \$10, up to \$50/week. Participants in the experimental group who did not meet or exceed their weekly weight goal had \$20 deducted from their accounts. No significant differences were found in baseline variables for either group; however, members of the control group lost an average of 5.9 pounds, whereas members of the experimental group lost an average of 18.1 pounds.

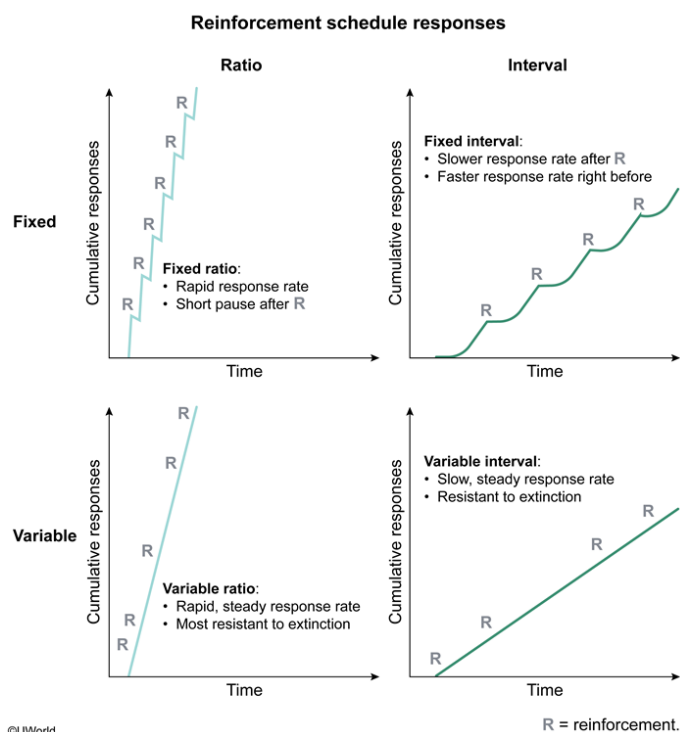
Which of the following behavioral patterns is most expected of participants in the Study 2 experimental group? Participants are most diligent with dieting and exercise:

- ☐ A. just after earning money. [20%]
- ☐ B. just after losing money. [23%]
- ✓ ☒ C. just before weekly weigh-ins. [47%]
- ☐ D. consistently throughout the week. [8%]

Correct

47% answered correctly

## Explanation:



Each schedule of reinforcement produces characteristic behavioral response patterns. The **ratio schedules**, which provide reinforcement after a consistent (fixed) or inconsistent (variable) number of behavioral responses, both produce **rapid response rates**. The **interval schedules**, which provide reinforcement after a consistent (fixed) or inconsistent (variable) amount of time, both produce **slower response rates**.

Fixed interval schedules (eg, providing a weekly weight-loss incentive) tend to produce increased behavioral responses *just before the reward* and a decreased pattern of responding just after the reward is received. For example, people tend to work harder and faster at the end of the week, just before receiving their weekly paychecks, and tend to be slower and more sluggish at the beginning of the week, after receiving their weekly paychecks.

Therefore, participants in the Study 2 experimental group who are on a fixed interval schedule should demonstrate similar behaviors. They should be most diligent with their diet and exercise regimens just before their weekly weigh-ins and least diligent with dieting and exercise right after the weekly weigh-in, regardless of whether they have earned or lost money **(Choices A and B)**.

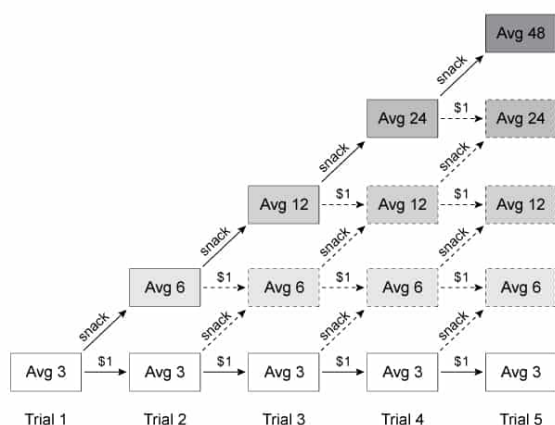
**(Choice D)** Variable interval schedules of reinforcement tend to produce slow and steady behavioral responses because the subject has no idea when the next reward is coming and the reward is not linked to behavior. For example, people tend to check email sporadically but consistently throughout the day (a slow and steady rate), because new email (a reinforcer) is received at unpredictable times that do not correspond to how often a person checks for email.

### **Educational objective:**

Reinforcement schedules produce characteristic behavioral response patterns: Ratio schedules produce rapid response rates, whereas interval schedules produce slower response rates. Fixed interval reinforcement schedules tend to produce increased behavior just before the reward and decreased behavior just after.

Researchers conducted a pair of studies to explore how rewards influence motivation and weight loss for those who are overweight or obese.

For Study 1, 20 overweight and 20 average-weight adults were invited to play a computer game consisting of a simple matching task in which participants earned a reward for getting multiple correct responses. Participants were allowed to select their own reward: Either \$1.00 or a 200-calorie vending machine snack (valued at \$1.00), such as a candy bar or bag of chips. For the first trial, participants were rewarded on an average of three correct responses (AVG3). Participants were then asked to choose between the food and monetary rewards. When a food reward was selected, the average number of correct responses required to earn the next reward doubled, but when a monetary reward was selected, the average number of correct responses required to earn the next reward remained the same (Figure 1). Results indicated that although there was no significant difference between the groups with regard to baseline self-reporting of hunger prior to the start of the computer tasks, overweight participants were five times more likely to choose the food rewards than average-weight participants.



**Figure 1** Study 1 reward schedule

For Study 2, obese participants were recruited for a 16-week weight-loss study and randomly assigned to one of two groups: A control group ( $n = 32$ ) or an experimental group ( $n = 35$ ). Control participants engaged in weekly weigh-ins only. Experimental participants engaged in weekly weigh-ins but also earned or lost money (put into an account and paid at the end of the study) based on their weigh-in goal, calculated each week based on current body mass index (BMI), age, and sex. Those who reached their weekly weight goal earned \$10; for each additional pound lost, participants earned another \$10, up to \$50/week. Participants in the experimental group who did not meet or exceed their weekly weight goal had \$20 deducted from their accounts. No significant differences were found in baseline variables for either group; however, members of the control group lost an average of 5.9 pounds, whereas members of the experimental group lost an average of 18.1 pounds.

Both studies use which of the following?

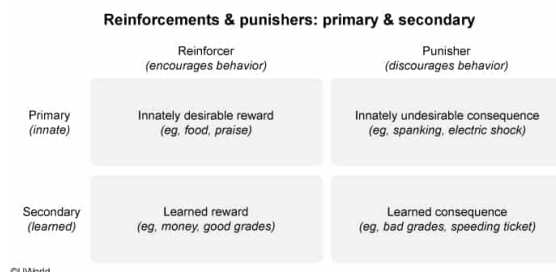
- ☐ A. Primary reinforcement [38%]

- ✓ ☒ B. Secondary reinforcement [58%]
- ☐ C. Primary punishment [1%]
- ☐ D. Secondary punishment [2%]

Correct

58% answered correctly

## Explanation:



In operant conditioning, reinforcers encourage behavior whereas punishers discourage behavior. Reinforcers and punishers can be primary (innate) or secondary (learned).

**Primary reinforcers** fulfill a biological urge and are naturally rewarding. Examples of primary reinforcers are food, comforting physical contact, and praise. **Secondary reinforcers** are not inherently rewarding but are *conditioned to be desirable* through associative learning. For example, if a child brings home good grades, the parent may reinforce this behavior with praise or candy (primary reinforcers) so the child feels happy. The child then associates feeling good with good grades, resulting in grades becoming a secondary reinforcer.

**Primary punishers** are naturally undesirable and cause negative emotions such as pain or fear. Examples of primary punishers include an electric shock or a loud noise. **Secondary punishers** are not inherently negative but are conditioned to be undesirable through associative learning. For example, if a child brings home bad grades, the parent may punish this behavior with a scolding (primary punisher) so the child feels bad. The child then associates feeling bad with bad grades, resulting in grades becoming a secondary punisher.

Both studies use money, a secondary reinforcer, as a reward.

**(Choice A)** Study 1 uses a food reward as a primary reinforcer. Study 2 does not include any primary reinforcement.

**(Choice C)** Neither study employs a primary punisher, such as a painful or fear-inducing stimulus.

**(Choice D)** Study 1 does not use any form of punishment. Study 2 uses the removal of a desirable stimulus (money) as a form of punishment, but this is referred to as negative punishment, not as a secondary punisher.

**Educational objective:**

Reinforcers encourage behavior; punishers discourage behavior. Reinforcers and punishers can be primary (innate) or secondary (learned). Primary reinforcers are innately desirable (eg, food), whereas secondary reinforcers are made desirable through learning (eg, good grades). Primary punishers are innately undesirable (eg, electric shock), whereas secondary punishers are made undesirable through learning (eg, bad grades).

**Time spent:**0 sec

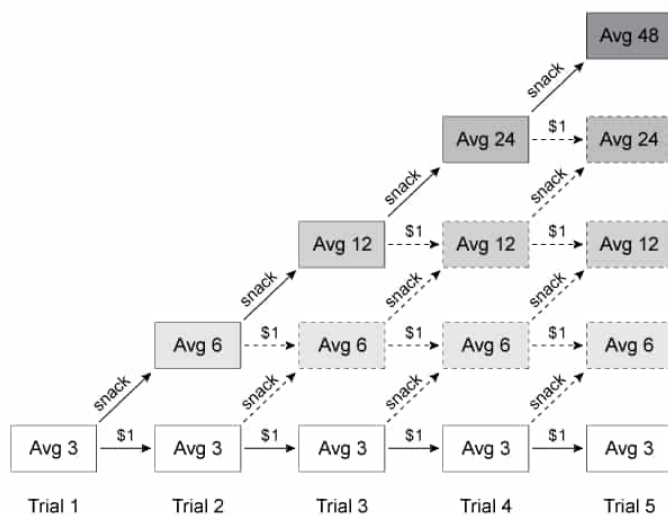
**Subject:**  
Behavioral Sciences

**QID:**400549

**System:**  
Learning, Memory, and Cognition

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**Figure 1** Study 1 reward schedule

For Study 2, obese participants were recruited for a 16-week weight-loss study and randomly assigned to one of two groups: A control group ( $n = 32$ ) or an experimental group ( $n = 35$ ). Control participants engaged in weekly weigh-ins only. Experimental participants engaged in weekly weigh-ins but also earned or lost money (put into an account and paid at

the end of the study) based on their weigh-in goal, calculated each week based on current body mass index (BMI), age, and sex. Those who reached their weekly weight goal earned \$10; for each additional pound lost, participants earned another \$10, up to \$50/week. Participants in the experimental group who did not meet or exceed their weekly weight goal had \$20 deducted from their accounts. No significant differences were found in baseline variables for either group; however, members of the control group lost an average of 5.9 pounds, whereas members of the experimental group lost an average of 18.1 pounds.

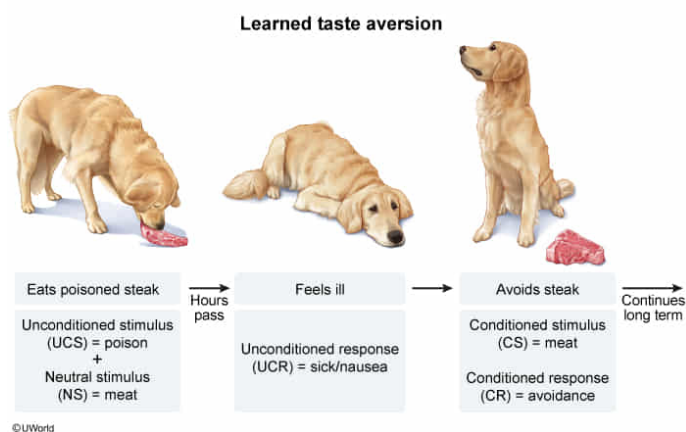
Suppose one of the participants in Study 1 earned several candy bars and became very ill after consuming them. This participant would most likely:

- ☐ A. avoid participating in any future research studies. [2%]
- ☐ B. avoid playing computer games for awhile until she feels better. [2%]
- ✓ ☒ C. develop a long-lasting aversion to the specific candy bars she consumed. [56%]
- ☐ D. develop a short-term aversion to the candy bars available during the study. [38%]

Correct

56% answered correctly

## Explanation:



A learned **taste aversion** is a specific and powerful type of classical conditioning that occurs when an organism becomes ill after consuming something (eg, food, alcohol). Whatever was consumed prior to the illness

becomes associated with the illness (even if it did not *cause* the illness) and is avoided by the organism afterward.

Unlike other forms of typical classical conditioning, which require two stimuli to be paired together repeatedly before the organism learns to associate the two, a taste aversion develops after just one pairing. In other words, an organism needs to become ill only once to associate the food consumed with the illness.

Similarly, whereas typical classical conditioning requires the two stimuli to be presented within a very short time frame for the organism to learn to associate them, taste aversions occur despite hours passing between the consumption of a food and subsequent illness.

Finally, whereas typical classical conditioning rapidly extinguishes when the two stimuli are no longer paired, taste aversions have long durations. In other words, once becoming ill, the organism will likely never consume the associated food again.

**(Choice A)** Although it is possible that the research participant might associate her participation in the study with becoming ill, she is *most likely* to associate her illness with the food, not the study or studies in general.

**(Choice B)** It is unlikely that the participant will associate the illness with the computer game. Taste aversions almost always link illness with foods (or smells), which is thought to be an evolutionary adaptation.

**(Choice D)** Taste aversions are long-enduring associations; they are not short-term.

### **Educational objective:**

Taste aversion is a specific and powerful type of classical conditioning that occurs after just one instance of becoming ill following the consumption of something. Taste aversions can develop despite many hours passing between consumption and illness and are long-lasting.

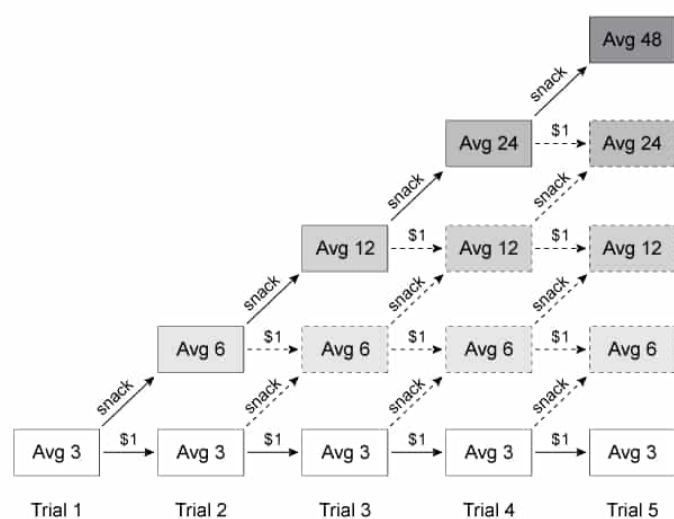
**Time spent:**0 sec  
**Subject:**  
Behavioral Sciences

**QID:**400550  
**System:**  
Learning, Memory, and Cognition



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**Figure 1** Study 1 reward schedule

For Study 2, obese participants were recruited for a 16-week weight-loss study and randomly assigned to one of two groups: A control group ( $n = 32$ ) or an experimental group ( $n = 35$ ). Control participants engaged in weekly weigh-ins only. Experimental participants engaged in weekly weigh-ins but also earned or lost money (put into an account and paid at

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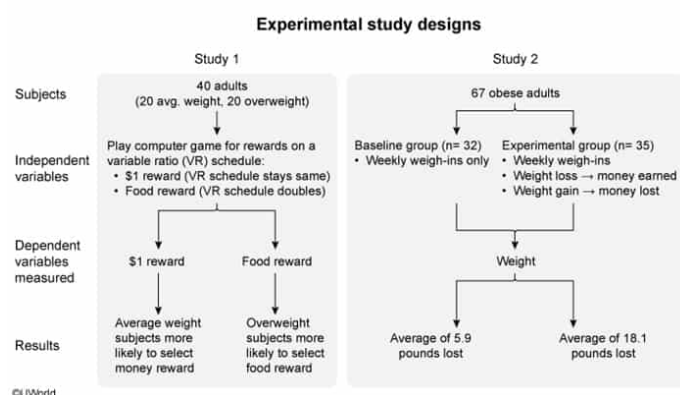
The participants' motivational state:

- ✔ ☒ A. was controlled for in Study 1 using self-report measures. [42%]
- ☐ B. was used to assign participants into groups in Study 2. [3%]
- ☐ C. appears to have been higher in the experimental group in Study 1. [20%]
- ☐ D. appears to have been correlated to BMI in Study 2. [32%]

Correct

43% answered correctly

## Explanation:



The purpose of Study 1 was to assess how weight corresponds to the type of reward a subject selects. The participants received rewards for correct answers during a computer game, with subsequent rewards becoming more challenging to earn for those who selected the food reward. The researchers manipulated the reward schedule (independent variable) to measure selection of the food or money reward (dependent variable).

Hunger might impact whether a participant was more likely to select a food reward, so hunger level was a potential **confounding variable**, one that might **affect the results** if not controlled for or eliminated by the researcher. Therefore, motivational state was controlled for by ensuring that levels of hunger were not significantly different among participants, assessed using self-report measures. If hunger levels had not been the same, any findings could be attributed to varying baseline levels of hunger rather than BMI.

**(Choice B)** The purpose of Study 2 was to assess how earning or losing money (independent variables) might impact weight loss (dependent variable). Therefore, if motivation were used to assign participants to the control or experimental group, this would result in groups that were fundamentally different, with results that could not be compared. To compare results, participants need to be assigned randomly (as they were in Study 2), such that average motivation levels between groups are similar.

**(Choice C)** Study 1 did not contain an experimental and control group; all participants were given a choice of reward, a decision that was correlated to BMI.

**(Choice D)** Motivation was not directly assessed in either study, although it could be inferred that participants in Study 2 were motivated by monetary rewards to lose weight. However, because obese participants were randomly assorted into two groups, it is not possible to correlate BMI with motivation, as the average BMI of each group should be comparable.

### **Educational objective:**

Potential confounding variables, such as motivation, must be controlled for at the outset of a study. Subjective states like hunger can be measured. If the groups do not vary in terms of the potential confounding variable, then any significant differences may be attributed to the independent variable(s).

**Time spent:**0 sec  
**Subject:**  
Behavioral Sciences

**QID:**400551  
**System:**  
Learning, Memory, and Cognition